AI-Powered Customer Lifetime Value Prediction System

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Problem Statement: Why Predict Customer Lifetime Value?

Strategic Challenge

Businesses struggle to identify which customers will generate the most long-term revenue, making it difficult to prioritise resources effectively.

Inefficient Spend

Traditional marketing treats all customers equally, leading to wasted budget on low-value customers whilst missing high-value opportunities.

Our Solution

Use AI to predict each customer's lifetime value to optimise marketing, retention, and resource allocation with precision.

Time Constraint

Build a fast, accurate, scalable CLV prediction system within 6 hours during our hackathon sprint.

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Key Features Included in Our System

01

Data-Driven CLV Prediction

Using historical purchase data and behavioural metrics including Recency, Frequency, and Monetary value analysis.

02

Machine Learning Model

XGBoost regressor trained to forecast future customer value with approximately 74% accuracy (R^2 score).

03

Customer Segmentation

Classify customers into high, medium, and low-value tiers for targeted marketing campaigns.

04

Interactive Dashboard

Visualise top customers, segments, and predicted CLV for actionable business insights.

05

Automated Pipeline

Data preprocessing, model training, and prediction integrated for quick iteration and deployment.

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Features Excluded or Deferred

1. **Real-Time Data Integration**

   Streaming data pipelines deferred due to time constraints .

2. **Advanced Churn Prediction**

   Survival analysis models and sophisticated churn prediction not implemented in this sprint.

3. **Deep Learning Approaches**

   LSTM networks for sequential purchase pattern analysis reserved for future development.

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How Our CLV Prediction System Works: Customer Segmentation

Our AI-powered system segments customers into four distinct categories based on their predicted Customer Lifetime Value (CLV), enabling highly targeted strategies for each group.

Champions (22%)

Our most valuable and loyal customers who generate significant long-term revenue. Focus on retention and premium offers.

Loyal (31%)

Regular purchasers with good CLV potential. Nurture these customers with relevant promotions to elevate them to Champions.

At Risk (28%)

Customers whose purchasing activity or engagement is declining. Implement re-engagement campaigns and special incentives.

Lost (19%)

Customers who have ceased purchasing or engaging. Focus on win-back strategies or re-evaluating acquisition costs.

This segmentation provides actionable insights, allowing businesses to allocate resources efficiently and tailor marketing efforts to maximise customer value across the entire base.

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Impact & Competitive Edge: Why Our Prototype Wins

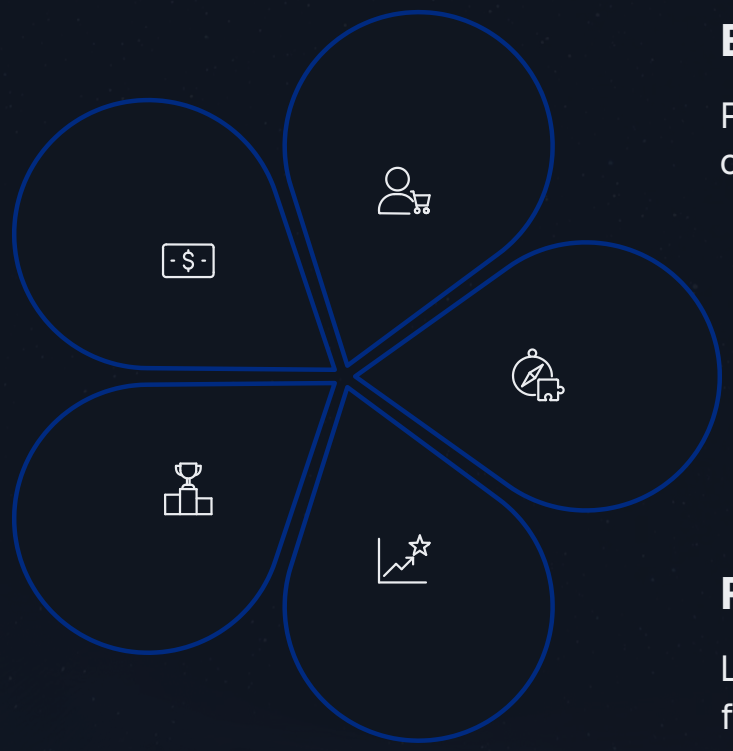
Our CLV prediction system isn't just about data; it's about transforming business strategy. By accurately forecasting customer value, we unlock significant competitive advantages.

Optimised ROI

Maximise return on marketing and sales investments by targeting high-value segments.

Market Leader

Gain a decisive edge over competitors with superior customer intelligence and personalised engagement.



Enhanced Retention

Proactively identify and re-engage at-risk customers to boost long-term loyalty.

Strategic Decisions

Empower business leaders with data-driven insights for smarter resource allocation.

Rapid Deployment

Leverage an agile, scalable system built for quick integration and fast results.

This prototype provides a powerful tool for businesses to not only understand their customers better but also to act decisively, driving growth and profitability in a competitive landscape.

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Team Roles & Next Steps

Data Engineer

Data cleaning and pipeline automation

ML Engineer

Model development and tuning

Data Analyst

Feature engineering and segmentation logic

Visualisation Expert

Dashboard creation and UX design

Project Lead

Integration, presentation, and demo coordination

Next Steps

Expand model features, integrate real-time data streams, and deploy for live marketing optimisation in production environments.

Thank you!

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